

Olist E-Commerce Analytics Report

Project Title: Olist E-Commerce Analytics & Performance Report

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Tool Used: SQL Server (SSMS), Power BI (Power Query, Data Modeling, DAX, Dynamic Dashboards & AI Visuals)

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1. Executive Summary

- **Project Overview:** A brief introductory paragraph about analyzing Olist e-commerce dataset (Brazilian market) to help management understand sales trends, delivery bottlenecks, and customer satisfaction factors.
- **Core Business Goals:**
 - Identify sales trends across product categories and regions.
 - Analyze order fulfillment and logistics delay impact.
 - Understand customer review score drivers.

2. Technical Workflow & Data Preparation Pipeline

A. SQL Server Analytics & Database Extraction

Database Exploration: Loading raw Olist relational database tables into SQL Server Management Studio.

Business Questions Answered via SQL

Monthly Sales Trend Over Time

```
10 SELECT
11 YEAR(O.order_purchase_timestamp) AS order_year,
12 MONTH(O.order_purchase_timestamp) AS order_month,
13 SUM(OI.price) AS total_sales
14 FROM olist_orders_dataset O
15 INNER JOIN olist_order_items_dataset OI
16 ON O.order_id = OI.order_id
17 GROUP BY YEAR(O.order_purchase_timestamp),
18 MONTH(O.order_purchase_timestamp)
19 ORDER BY order_year, order_month;
```

00 % No issues found

order_year	order_month	total_sales
2016	9	267.360008239746
2016	10	49507.659778595
2016	12	10.8999996185303
2017	1	120312.869802952
2017	2	247303.019815922
2017	3	374344.299495697
2017	4	359927.229712963
2017	5	506071.139301777
2017	6	433038.600114107
2017	7	498031.480387211
2017	8	573971.680464983
2017	9	634401.680417362

Top 10 Products by Total Sales vs Avg Price

```
25 SELECT TOP 10
26 OC.product_category_name_english AS product_category,
27 ROUND(SUM(OI.price), 2) AS total_sales,
28 ROUND(AVG(OI.price), 2) AS avg_selling_price
29 FROM olist_products_dataset P
30 LEFT JOIN olist_order_items_dataset OI
31 ON P.product_id = OI.product_id
32 LEFT JOIN olist_order_category_dataset OC
33 ON OC.product_category_name = P.product_category_name
34 GROUP BY OC.product_category_name_english
35 ORDER BY total_sales desc;
```

% No issues found

product_category	total_sales	avg_selling_price
health_beauty	1258681.34	130.16
watches_gifts	1205005.68	201.14
bed_bath_table	1036988.68	93.3
sports_leisure	988048.97	114.34
computers_accessories	911954.32	116.51
furniture_decor	729762.49	87.56
cool_stuff	635290.85	167.36
housewares	632248.66	90.79
auto	592720.11	139.96
garden_tools	485256.46	111.63

B. Power Query Cleaning & Transformation

Data Hygiene: Handled missing values, removed duplicate records, and ensured proper data types for all numerical, date, and currency fields.

Data Standardization: Cleaned text structures and standardized datetime fields to support seamless time-intelligence functions in Power BI.

C. Data Modeling & Schema Design

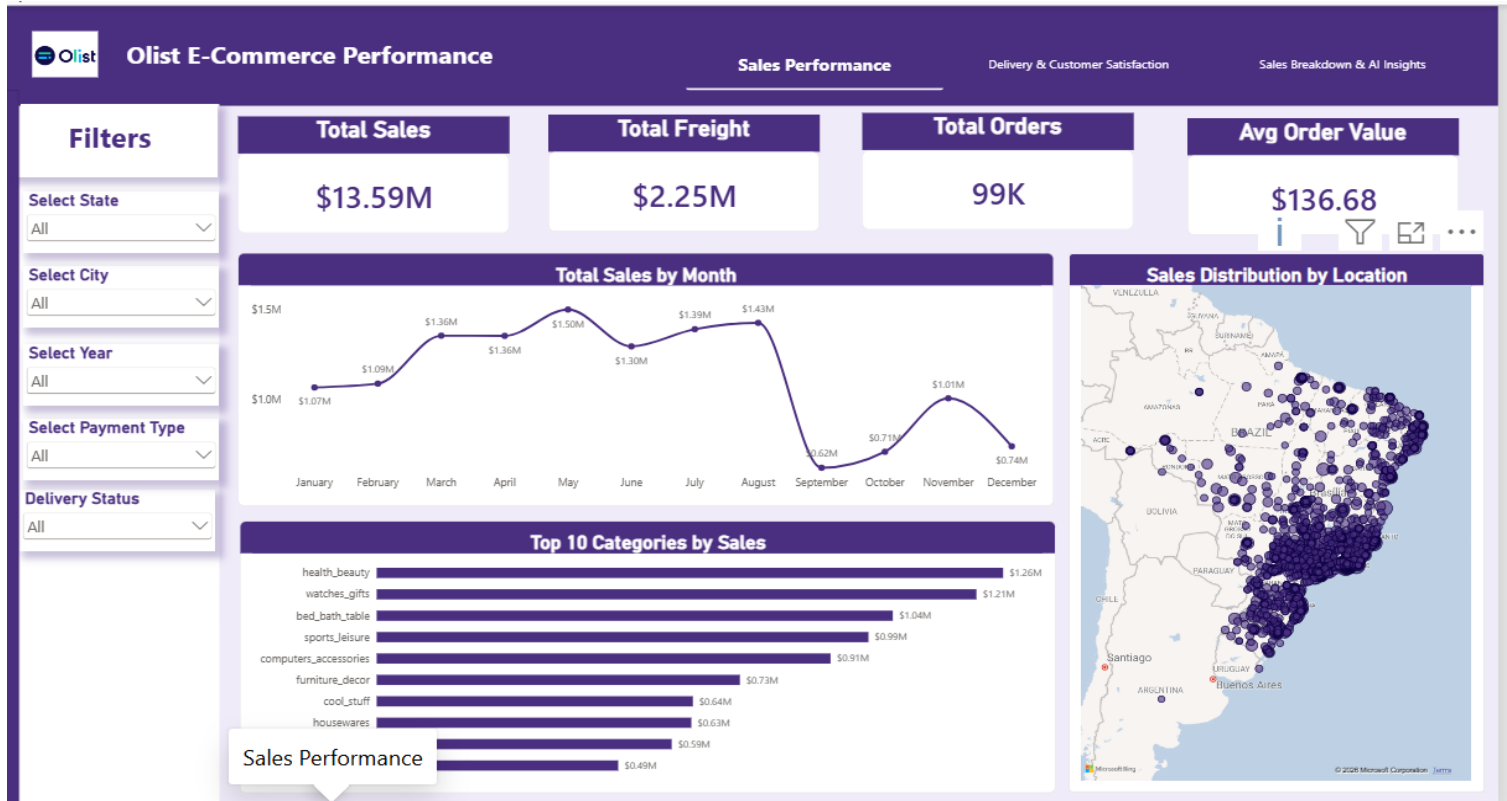
Geolocation Conflict Resolution: Addressed non-unique ZIP code prefixes within the geolocation dataset to prevent ambiguous data relationships.

Role-Playing Dimensions: Created separate Customer Geolocation and Seller Geolocation tables to build clear 1-to-Many relationships without circular dependencies.

Star Schema Implementation: Designed an optimized Star Schema connecting Fact tables (Orders, Order Items) with Dimension tables (Customers, Sellers, Products, Reviews, Payments)

D. Power BI Dashboard Development

Page 1: Sales Performance



• Key Insights:

- **Total Revenue & Volume:** The platform generated \$13.59M in total sales across 99K orders, with an Average Order Value (AOV) of \$136.68 and \$2.25M in total freight costs.

- **Revenue Seasonality:** Sales peaked significantly between May and August (reaching ~\$1.5M per month), followed by a sharp decline in September and a temporary spike in November.

- **Annual Sales Trajectory (Partial Data):** Across the recorded periods, sales showed rapid expansion, moving from \$0.05M during late 2016 (3 months), surging to \$6.2M in 2017 (full year), and reaching \$7.4M through September 2018 (9 months), indicating continuous platform scaling.

- **Quarterly Seasonality:** Demand peaks during Q2 (\$4.2M), driven by major promotional mid-year events, before gradually cooling down in Q3 (\$3.4M) and Q4 (\$2.5M).

- **Category Leaders:** The "health_beauty" category is the top revenue driver (\$1.26M), closely followed by "watches_gifts" (\$1.21M) and "bed bath_table" (\$1.04M).

- **Geographic Concentration:** Sales density is heavily concentrated in the Southeastern states of Brazil, particularly around major urban centers.

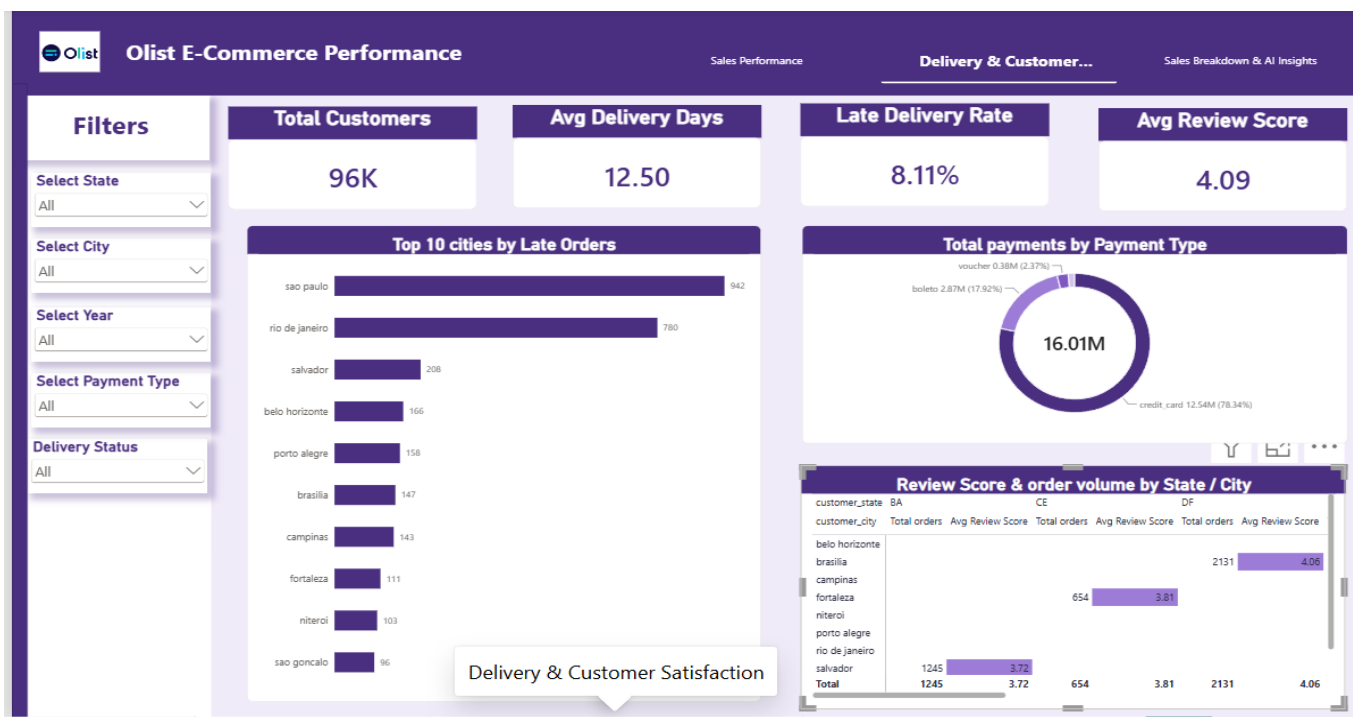
- **Strategic Recommendations:**

- **Targeted Promotions:** Increase promotional campaigns and stock availability between April and August to capitalize on the high-demand season.

- **Freight Optimization:** Since shipping costs account for nearly 16.5% of total sales (\$2.25M), negotiate bulk shipping contracts in top-concentrated regions to reduce delivery costs and increase customer retention.

- **Cross-Selling Strategies:** Bundle lower-performing items with top categories like "health beauty" and "watches gifts" to boost average order value further.

Page 2: Delivery & Customer Satisfaction



- **Key Insights:**

- **Fulfillment Efficiency:** The average delivery time across all orders is 12.50 days, with an overall Late Delivery Rate of 8.11%.

- **Customer Satisfaction:** Despite logistics challenges, the platform maintains a healthy Average Review Score of 4.09 out of 5.

- **Delay Hotspots:** Late deliveries are heavily concentrated in major metropolitan hubs, with São Paulo and Rio de Janeiro recording the highest volume of delayed orders.

- **Payment Preferences:** Credit Card is the dominant payment method, accounting for 78.34% of total payment volume (\$16.01M), followed by Boleto.

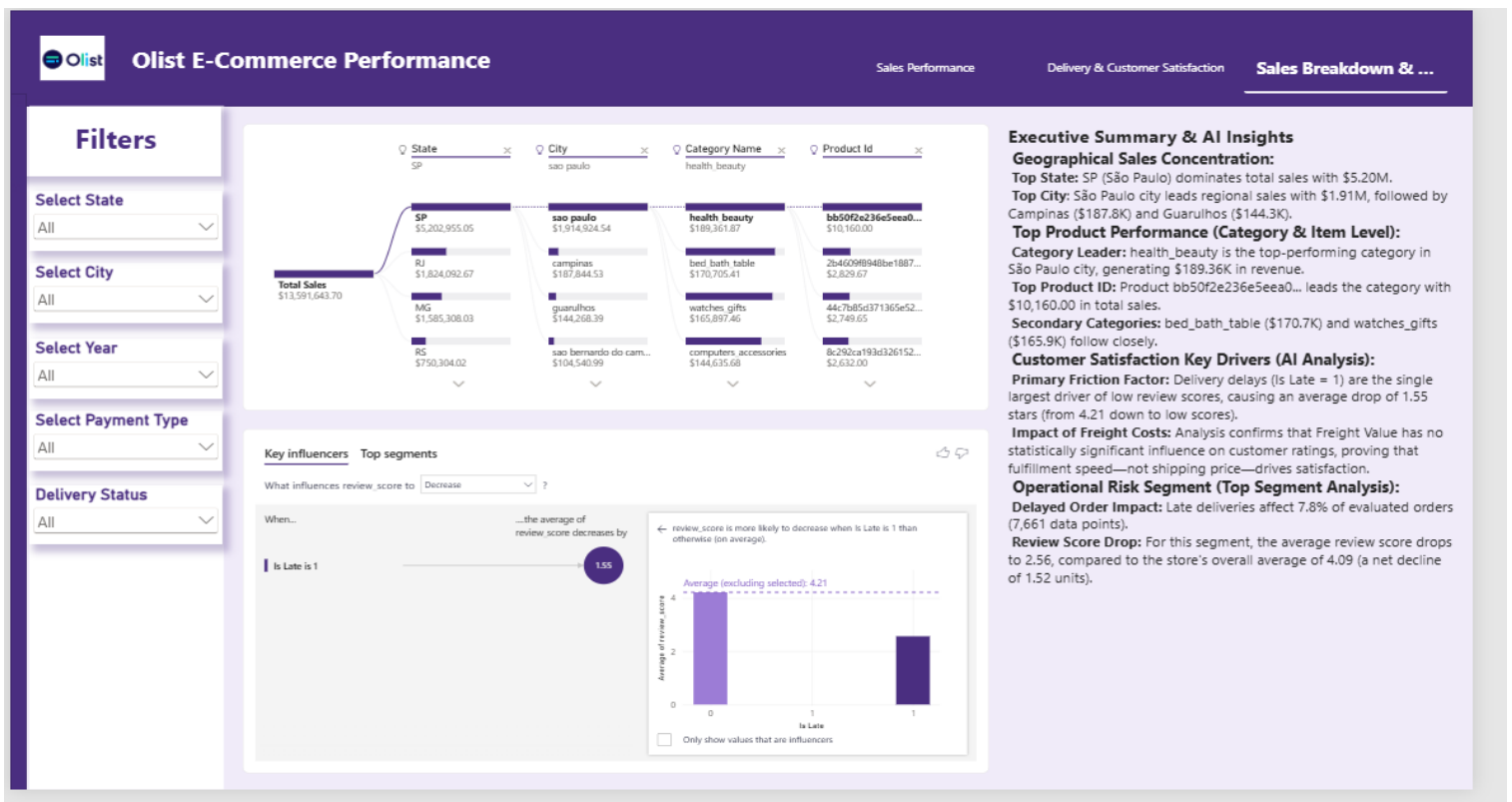
- **Strategic Recommendations:**

- **Local Warehouse Expansion:** Establish regional fulfillment centers or partner with local 3PL providers near São Paulo and Rio de Janeiro to shorten shipping times and reduce the 8.11% late delivery rate.

- **Seller Logistics SLA:** Enforce stricter dispatch SLAs for sellers in high-delay regions to improve overall customer experience and protect review scores.

- **Flexible Payment Incentive:** Offer targeted discounts or cashback for instant payment methods (like credit card or digital wallets) to minimize payment processing delays associated with Boleto.

Page 3: Sales Breakdown & AI Insights



- **Key Insights:**

- **Geographical Sales Concentration:** Sales are heavily localized, with São Paulo (SP) dominating total revenue at \$5.20M. São Paulo city leads all urban centers with \$1.91M, followed by Campinas (\$187.8K) and Guarulhos (\$144.3K).

- **Product-Level Performance:** Within the top-performing "health_beauty" category in São Paulo (\$189.36K), the leading Product ID (bb50f2e236e5eea0...) generated \$10.16K in sales.

- **Primary Friction Factor (AI Analysis):** Advanced Key Influencers analysis identifies delivery delays (Is Late = 1) as the primary driver of low review scores, triggering a significant average drop of 1.55 stars (reducing average rating from 4.21 down to 2.66).

- **Freight Cost Neutrality:** AI analysis proves that freight cost has no statistically significant impact on review scores, confirming that delivery speed—rather than shipping price—directly drives customer satisfaction.

- **Strategic Recommendations:**

- **Targeted Delay Mitigation:** Focus operational interventions exclusively on logistics velocity rather than shipping subsidies, as delay is the sole metric damaging review scores.

- **Priority Inventory Allocation:** Optimize product availability and warehouse storage for top-performing SKUs in high-demand hubs like São Paulo to prevent stockouts and speed up dispatch.